10/17/2026

Math 231A Lecture Notes:

I have a few more notes about harmonic functions to take and then I will move on to talking about the heat equation. To start off, the following is useful for bounding harmonic functions. (Also once again I'll just be assuming $n \geq 2$ always.)

Harnack's Inequality: Suppose $U \subseteq \mathbb{R}^n$ is an open set and K is a connected compact set contained in U. Then there exists a constant $C > 0$ such that:

$$\max_{x \in K}(u) \leq C \min_{x \in K}(u)$$

for all harmonic functions $u : U \to \mathbb{R}^n$ with $u \geq 0$ on U.

Note that C does not depend on the function u at all.

Proof:
Set $r = \frac{1}{4}d(K, \partial U) := \frac{1}{4}\inf\{|x - y| : x \in K, y \in \partial U\}$. Since K and ∂U are disjoint closed sets with K compact, we know $r > 0$.

Now for any $x \in K$ and $y \in \overline{B_r(x)}$ we know by the mean value theorem plus the fact that u is nonnegative that:

$$u(x) = \frac{1}{(2r)^n \alpha(n)} \int_{B_{2r}(x)} u \geq \frac{1}{(2r)^n \alpha(n)} \int_{B_r(y)} u = \frac{r^n \alpha(n)}{(2r)^n \alpha(n)} u(y) = \frac{1}{2^n} u(y)$$

But now observe that because K is compact, we can cover cover K with a finite number N of balls: $B^{(1)}, \ldots, B^{(N)}$, each with radius $r/2$ and centered at a point in K. Next, I claim that because K is connected, we can for any $x, y \in K$ always find $z_1, z_2, \ldots, z_\ell \in K$ (where $\ell \leq 2N - 2$) such that $|x - z_1| \leq r$, $|z_i - z_{i+1}| \leq r$ for all i, and $|z_\ell - y| \leq r$. In fact, I'll even prove a slightly more general lemma.

Lemma: Suppose (X, d) is any metric space, $A \subseteq X$ is a connected set, and there exists finitely many sets $E_1, \ldots, E_N$ of diameter at most r covering A. Given any $x, y \in A$ we can always find points $z_1, \ldots, z_\ell \in A$ (where $\ell \leq 2N - 2$) such that $d(x, z_1) \leq r$, $d(z_i, z_{i+1}) \leq r$ for each i, and $d(z_\ell, y) \leq r$.

Proof:
It suffices to fix $x \in A$. Now to start off, I shall disregard getting an upper bound for ℓ and just show that for any $y \in A$ there even exists points $z_1, \ldots, z_\ell \in A$ such that $d(x, z_1) \leq r$, $d(z_i, z_{i+1}) \leq r$ for each i, and $d(z_\ell, y) < r$. To do this, I shall let $C \subseteq A$ be the set of all $y \in A$ such that this is possible. Note that C is nonempty because trivially $x \in C$. Also, we can easily verify both C and $A - C$ are open subsets of A. Since A is connected, this thus proves that $C = A$.

Now fixing $y \in A$, we've shown that there are $z_1, \ldots, z_\ell \in A$ with $d(x, z_1) \leq r$, $d(z_i, z_{i+1}) \leq r$ for each i, and $d(z_\ell, y) \leq r$. For ease of notation, I'll let $z_0 := x$ and $z_{\ell+1} := y$.

If $\ell > 2N-2$ then we know by the pigeonhole principle that there are integers $0 \leq i_1 < i_2 < i_3 \leq \ell + 1$ with $z_{i_1}, z_{i_2}, z_{i_3}$ all in the same set E_j. Then as E_j has diameter at most r, we know $d(z_{i_1}, z_{i_3}) \leq r$. So, we can disregard each of the z_j with $i_1 < j < i_3$ to get another collection of points satisfying our lemma. In particular, this collection has at least one less element because we disregarded z_{i_2}.

By repeating the prior reasoning finitely many times, we will eventually get that $\ell \leq 2N-2$.

Based on the prior lemma, we can now see for any nonnegative harmonic function $u : U \to \mathbb{R}$ and any $x, y \in K$ that:

$$u(x) \geq \left(\tfrac{1}{2^n}\right)^{2N-1} u(y). \blacksquare$$

I also want to motivate why we'd care about Green's functions.

Firstly, I want to briefly go over how the textbook (see *citation 16*) that my 231 class is loosely based on defines what it means for an open set to have a C^r boundary (where $r \geq 1$).

Let $U \subseteq \mathbb{R}^n$ be an open set and let $k \geq 1$. Evans (the author of that textbook) says that U has a C^k boundary if for each $x^{(0)} \in \partial U$ there exists $r > 0$ and a C^k function $\gamma : \mathbb{R}^{n-1} \to \mathbb{R}$ such that upon relabeling and reorienting the coordinate axes if necessary, we have that:

$$U \cap B(x^{(0)}, r) = \{x \in B(x^{(0)}, r) : x_n < \gamma(x_1, \ldots, x_{n-1})\}$$

I claim that this happens if and only if U is the interior of a regular domain D without corners and whose boundary is ∂U.

$(\Longleftarrow)$

Hopefully it is obvious how this direction is an easy result of the theorem on *pages 1170-1171*.

$(\Longrightarrow)$

We know $D = U \cup \partial U$ is a closed subset of $\mathbb{R}^n$. So if it were a submanifold it would be properly embedded. Also it's clear that U is a C^r submanifold of codimension 0. At the same time, if $p \in \partial U$ and we let γ and $B(x^{(0)}, r)$ be as described in the above definition, then we can apply a change of coordinates:

$$(x_1, \ldots, x_n) \mapsto (x_1, \ldots, x_{n-1}, \gamma(x_1, \ldots, x_{n-1}) - x_n)$$

in order to get an n-dimensional half slice chart for D centered at p.

Now suppose $U \subseteq \mathbb{R}^n$ is a bounded open set with a C^1 boundary. At any point $p \in \partial U$ we let $v(p)$ denote the outward pointing unit normal vector. Also let $\Phi : \mathbb{R}^n - \{0\} \to \mathbb{R}$ be the fundamental solution of $-\triangle$. (See *page 1181*). And let $\tau_x f(y) \coloneqq f(y - x)$.

Observation: Let $u \in C^2(\overline{U})$ be an arbitrary function. Then for any $x \in U$:

$$u(x) = \int_{\partial U} \Phi(y-x)\frac{\partial u}{\partial v}(y) - u(y)\frac{\partial(\tau_x\Phi)}{\partial v}(y)\mathrm{d}S(y) - \int_U \Phi(y-x)\triangle u(y)\mathrm{d}y$$

Proof:
Fix any $x \in U$ and let $\varepsilon > 0$ be small enough so that $\overline{B_\varepsilon(x)} \subseteq U$. Then let $V_\varepsilon = U - B_\varepsilon$.

We have by Green's 2nd identity that:

$$\begin{aligned}\int_{V_\varepsilon} \Phi(y-x)\triangle u(y)\mathrm{d}y &= \int_{\partial V_\varepsilon} \Phi(y-x)\tfrac{\partial u}{\partial v}(y) - u(y)\tfrac{\partial(\tau_x\Phi)}{\partial v}(y)\mathrm{d}S(y)\\ &= \int_{\partial U} \Phi(y-x)\tfrac{\partial u}{\partial v}(y) - u(y)\tfrac{\partial(\tau_x\Phi)}{\partial v}(y)\mathrm{d}S(y)\\ &\quad - \int_{\partial B_\varepsilon(x)} \Phi(y-x)\tfrac{\partial u}{\partial v}(y) - u(y)\tfrac{\partial(\tau_x\Phi)}{\partial v}(y)\mathrm{d}S(y)\end{aligned}$$

But now $\left|\int_{\partial B_\varepsilon(x)} \Phi(y-x)\frac{\partial u}{\partial v}(y)\mathrm{d}S(y)\right| = \int_{\partial B_\varepsilon(x)} |\Phi(y-x)| \cdot |\nabla u(y)|\mathrm{d}S(y)$. For small enough ε we can guarentee that there is some $C > 0$ with $|\nabla u(y)| \leq C$ for all $y \in B_\varepsilon(x)$. And thus:

$$\left|\int_{\partial B_\varepsilon(x)} \Phi(y-x)\tfrac{\partial u}{\partial v}(y)\mathrm{d}S(y)\right| \leq C\varepsilon^{n-1}n\alpha(n)\max_{y\in\partial B_\varepsilon(0)}|\Phi(y)|$$

If $n = 2$ then $\varepsilon\ln(\varepsilon) \to 0$ as $\varepsilon \to 0$. In other words $\ln(\varepsilon) = o(\varepsilon^{-1})$ as $\varepsilon \to 0$.
Similarly, if $n \geq 3$ then $\varepsilon^{2-n} = o(\varepsilon^{1-n})$ as $\varepsilon \to 0$. So in general, we can conclude:

$$C\varepsilon^{n-1}n\alpha(n)\max_{y\in\partial B_\varepsilon(0)}|\Phi(y)| = \varepsilon^{n-1}o(1-n) = o(1) \text{ as } \varepsilon \to 0.$$

Meanwhile, note that if $n = 2$ and $y \neq 0$ then $\nabla\Phi(y)\cdot\frac{y}{|y|} = \frac{-1}{2\pi|y|}$. Similarly, if $n \geq 3$ then $\nabla\Phi(y)\cdot\frac{y}{|y|} = \frac{-1}{n\alpha(n)|y|^{n-1}}$. So, we can say that:

$$\int_{\partial B_\varepsilon(x)} u(y)\tfrac{\partial(\tau_x\Phi)}{\partial v}(y)\mathrm{d}S(y) = \tfrac{-1}{n\alpha(n)\varepsilon^{n-1}}\int_{\partial B_\varepsilon(x)} u(y)\mathrm{d}S(y) \to -u(x) \text{ as } \varepsilon \to 0.$$

And thus we've proven that by taking $\varepsilon \to 0$, we have that:

$$\begin{aligned}\int_{V_\varepsilon} \Phi(y-x)\triangle u(y)\mathrm{d}y &= \int_{\partial V_\varepsilon} \Phi(y-x)\tfrac{\partial u}{\partial v}(y) - u(y)\tfrac{\partial(\tau_x\Phi)}{\partial v}(y)\mathrm{d}S(y)\\ &= \int_{\partial U} \Phi(y-x)\tfrac{\partial u}{\partial v}(y) - u(y)\tfrac{\partial(\tau_x\Phi)}{\partial v}(y)\mathrm{d}S(y) - u(x). \blacksquare\end{aligned}$$

Based on the prior observation, we can thus say that u is uniquely determined on U by u and $\frac{\partial u}{\partial v}$ on ∂U as well as $\triangle u$ on U. That said though, our PDE is actually over-determined if we give boundary conditions for both u and $\frac{\partial u}{\partial v}$ on ∂U.

We focus on the Dirichlet problem (i.e. where we don't control for $\frac{\partial u}{\partial v}$ at all). So, we need to cancel out the integral $\int_{\partial U} \Phi(y-x)\frac{\partial u}{\partial v}(y)\mathrm{d}S(y)$. To do this, we shall assume there exists a corrector function $\phi : U \times \overline{U} \to \mathbb{R}$ such that:

- $\triangle_y\phi(x,y) = 0$ for all $x, y \in U$
- $\phi(x,y) = \Phi(y-x)$ for all $x \in U$ and $y \in \partial U$.
- $\phi^x(y) \coloneqq \phi(x,y)$ satisfies that $\phi^x \in C^2(\overline{U})$.

  (Don't worry for now about if ϕ exists. It will be our goal later to prove it exists and calculate it in the cases we are interested in.)

By Green's 2nd identity we have that:

$$-\int_U \phi^x(y)\triangle u(y)\mathrm{d}y = \int_{\partial U} u(y)\frac{\partial \phi^x}{\partial v}(y) - \Phi(y-x)\frac{\partial u}{\partial v}(y)\mathrm{d}S(y).$$

And by combining that with our prior observation we get that:

$$u(x) = \int_{\partial U} u(y)\left[\frac{\partial \phi^x}{\partial v}(y) - \frac{\partial(\tau_x\Phi)}{\partial v}(y)\right]\mathrm{d}S(y) + \int_U \left[\phi^x(y) - \Phi(y-x)\right]\triangle u(y)\mathrm{d}y$$

Notably the prior expression writes u purely in terms of $u|_{\partial U}$ and $\triangle u$ on U.

We define Green's function associated to the domain $\overline{U}$ to be the map:

$$G(x,y) \coloneqq \Phi(y-x) - \phi^x(y).$$

And by the prior reasoning, we can conclude the following theorem:

Theorem: Suppose $u \in C^2(\overline{U})$ satisfies that $-\triangle u = f$ on U and $u = g$ on ∂U (where f and g are continuous). Then:

$$u(x) = -\int_{\partial U} u(y)\frac{\partial G}{\partial v}(x,y)\mathrm{d}S(y) + \int_U f(y)G(x,y)\mathrm{d}y.$$

(Here $\frac{\partial G}{\partial v}(x,y) = \nabla_y G(x,y)\cdot v$.)

Given an open set $U \subseteq \mathbb{R}^n$ with a C^1 boundary, another way we can interpret Green's function associated to $\overline{U}$ is that it is the map satisfying that $-\triangle_y G(x,y) = \delta_x$ for all $x, y \in U$ and $G(x,y) = 0$ for all $x \in U$ and $y \in \partial U$.

Theorem: For all $x, y \in U$ with $x \neq y$ we have that $G(x,y) = G(y,x)$.

Proof:

Fix $x, y \in U$ and define $w_1(z) \coloneqq G(x,z)$ and $w_2(z) \coloneqq G(y,z)$. Then $\triangle w_1(z) = 0$ for all $z \in U - \{x\}$, $\triangle w_2(z) = 0$ for all $z \in U - \{y\}$, and $w_1(z) = w_2(z) = 0$ if $z \in \partial U$. Now let $\varepsilon > 0$ be such that $B_{2\varepsilon}(x) \cap B_{2\varepsilon}(y) = \emptyset$ and $B_{2\varepsilon}(x) \cup B_{2\varepsilon}(y) \subseteq U$.

By Green's 2nd Identity applied to the domain $V = U - (B_\varepsilon(x) \cup B_\varepsilon(y))$ we have that:

$$0 = -\int_{\partial B_\varepsilon(x)\cup\partial B_\varepsilon(y)} w_1\frac{\partial w_2}{\partial v} - w_2\frac{\partial w_1}{\partial v}\mathrm{d}S.$$

Or after some rearranging, we can equivalently say that:

$$\int_{\partial B_\varepsilon(y)} w_1\frac{\partial w_2}{\partial v}\mathrm{d}S - \int_{\partial B_\varepsilon(y)} w_2\frac{\partial w_1}{\partial v}\mathrm{d}S = \int_{\partial B_\varepsilon(x)} w_2\frac{\partial w_1}{\partial v}\mathrm{d}S - \int_{\partial B_\varepsilon(x)} w_1\frac{\partial w_2}{\partial v}\mathrm{d}S$$

But now observe the following:

w_1 is a smooth function on $B_\varepsilon(y)$ (since it is harmonic on $U - \{x\}$). So, we can find a constant $C > 0$ with $|\nabla w_1| \leq C$ on $\overline{B_\varepsilon(y)}$. And then:

$$\left|\int_{\partial B_\varepsilon(y)} w_2\frac{\partial w_1}{\partial v}\mathrm{d}S\right| \leq C\int_{\partial B_\varepsilon(y)} |w_2|\mathrm{d}S \leq C\varepsilon^{n-1}n\alpha(n)\max_{z\in\partial B_\varepsilon(y)}|w_2(z)|$$

Next, $w_2(z) = \Phi(z-y) - \phi^y(z)$. But ϕ^y is a smooth (since it is harmonic) function around y. So there exists a constant $D > 0$ with $|\phi^y(z)| \leq D$ for all $z \in \overline{B_\varepsilon(y)}$.

And now:

$$C\varepsilon^{n-1}n\alpha(n)\max_{z\in\partial B_\varepsilon(y)}|w_2(z)|\leq CD\varepsilon^{n-1}n\alpha(n)+o(1)\text{ as }\varepsilon\to 0$$

This finishes proving that $\int_{\partial B_\varepsilon(y)} w_2\frac{\partial w_1}{\partial v}\mathrm{d}S\to 0$ as $\varepsilon\to 0$.

Analogous reasoning as in the prior observation shows that $\int_{\partial B_\varepsilon(x)} w_1\frac{\partial w_2}{\partial v}\mathrm{d}S\to 0$ as $\varepsilon\to 0$. And so, we can conclude that:

$$\lim_{\varepsilon\to 0^+}\int_{\partial B_\varepsilon(y)} w_1\frac{\partial w_2}{\partial v}\mathrm{d}S=\lim_{\varepsilon\to 0^+}\int_{\partial B_\varepsilon(x)} w_2\frac{\partial w_1}{\partial v}\mathrm{d}S$$

Finally, we just have to compute these two limits.

Note that:

$$\int_{\partial B_\varepsilon(y)} w_1\frac{\partial w_2}{\partial v}\mathrm{d}S=-\frac{1}{n\alpha(n)\varepsilon^{n-1}}\int_{\partial B_\varepsilon(y)} w_1\mathrm{d}S+\int_{\partial B_\varepsilon(y)} w_1\frac{\partial \phi^y}{\partial v}\mathrm{d}S.$$

As $\varepsilon\to 0$ we have that $-\frac{1}{n\alpha(n)\varepsilon^{n-1}}\int_{\partial B_\varepsilon(y)} w_1\mathrm{d}S\to -w_1(y)$.

Meanwhile, because w_1 and ϕ^y are smooth on $\overline{B_\varepsilon(y)}$ (since they are harmonic), we can find a constant C such that $|w_1\nabla\phi^y|\leq C$ on $\overline{B_\varepsilon(y)}$. And now:

$$\left|\int_{\partial B_\varepsilon(y)} w_1\frac{\partial \phi^y}{\partial v}\mathrm{d}S\right|\leq C\varepsilon^{n-1}\alpha(n)n\to 0\text{ as }\varepsilon\to 0.$$

This finishes showing that $\lim_{\varepsilon\to 0^+}\int_{\partial B_\varepsilon(y)} w_1\frac{\partial w_2}{\partial v}\mathrm{d}S=-w_1(y)$.

Analogous computations show $\lim_{\varepsilon\to 0^+}\int_{\partial B_\varepsilon(x)} w_2\frac{\partial w_1}{\partial v}\mathrm{d}S=-w_2(x)$

So, we've shown that $-G(x,y)=-w_1(y)=-w_2(x)=-G(y,x)$. ■

Right now I don't have the tools to prove when a Green's function does or does not exist. However, I'll at least construct Green's function explicitely for the domain $\overline{B_1(0)}$.

First I'll find the correction function ϕ^x when $x\neq 0$.

Given any $x\in\mathbb{R}^n-\{0\}$ let us denote $\widetilde{x}=\frac{x}{|x|^2}$. Importantly, if $x\in B_1(0)$ then $|\widetilde{x}|>1$. In turn, we know that $|x|(y-\widetilde{x})\neq 0$ for all $y\in\overline{B}_1(0)$. And in turn, upon letting $\phi^x(y)=\Phi(|x|(y-\widetilde{x}))$, we have that:

$$\triangle\phi^x(y)=|x|^2\triangle\Phi(|x|(y-\widetilde{x}))=|x|^2\cdot 0=0\text{ for all }y\in B_1(0).$$

Also note that if $|y|=1$ then:

$$|x|^2|y-\widetilde{x}|^2=|y|^2|x|^2-2x\cdot y+1=1|x|^2-2x\cdot y+|y|^2=|y-x|^2$$

So $|x||y-\widetilde{x}|=|y-x|$ and this proves that $\phi^x(y)=\Phi(|x|(y-\widetilde{x}))=\Phi(y-x)$ for all $y\in\partial B_1(0)$.

Meanwhile, if $x=0$ then just set the correction function equal to:

$$\phi^0\equiv\begin{cases}0 & \text{if } n=2\\ \frac{1}{n(n-2)\alpha(n)} & \text{if } n=3.\end{cases}$$

Constant functions are trivially harmonic and we know $\phi^0(y) = \Phi(y)$ trivially when $|y| = 1$.

With that, we are basically done. In the case that $n \geq 3$ we have that:

If $x \neq 0$ then:

$$G(x,y) = \Phi(y-x) - \Phi(|x|(y-\widetilde{x})) = \frac{1}{n(n-2)\alpha(n)}\left(\frac{1}{|y-x|^{n-2}} - \frac{1}{|x|^{n-2}|y-\widetilde{x}|^{n-2}}\right)$$

Meanwhile, $G(0,y) = \Phi(y) - \frac{1}{n(n-2)\alpha(n)} = \frac{1}{n(n-2)\alpha(n)}\left(\frac{1}{|y|^{n-2}} - 1\right)$.

As for the case $n = 2$, we have that:

If $x \neq 0$ then:

$$G(x,y) = \Phi(y-x) - \Phi(|x|(y-\widetilde{x})) = \frac{-1}{2\pi}\left(\ln|y-x| - \ln(|x|\cdot|y-\widetilde{x}|)\right)$$

Meanwhile, $G(0,y) = \Phi(y) - 0 = \frac{-1}{2\pi}\ln|y|$.

I'd like to use this computation to find an analog of the Poisson integral formula on *page 808*.

For all $n \geq 2$ and $1 \leq i \leq n$ we can calculate that $\frac{\partial}{\partial y_i}\left(\Phi(y-x)\right) = \frac{x_i - y_i}{n\alpha(n)|x-y|^n}$.

In a similar vein, if $x \neq 0$ and $n \geq 2$ then:

$$\frac{\partial}{\partial y_i}\left(\Phi(|x|(y-\widetilde{x}))\right) = \frac{1}{n\alpha(n)|x|^{n-1}|y-\widetilde{x}|^{n-1}} \cdot \frac{|x|}{2|y-\widetilde{x}|} \cdot 2(y_i - \frac{x_i}{|x|^2}) = \frac{1}{n\alpha(n)} \cdot \frac{x_i - y_i|x|^2}{|x|^n|y-\widetilde{x}|^n}$$

In particular, if $|y| = 1$ then $|x|^n|y-\widetilde{x}|^n = |y-x|^n$.

Therefore, we get for all $n \geq 2$, $1 \leq i \leq n$, $x \in B_1(0) - \{0\}$, and $y \in \partial B_1(0)$ that:

$$\frac{\partial}{\partial y_i}G(x,y) = \frac{-1}{n\alpha(n)} \cdot \frac{1-|x|^2}{|x-y|^n} \cdot y_i$$

Also, the prior result extends to when $x = 0$. After all, because the correction term is constant we know that:

$$\frac{\partial}{\partial y_i}G(x,y) = \frac{\partial \Phi}{\partial y_i}(y) = \frac{-1}{n\alpha(n)} \cdot \frac{1}{|y|^n} \cdot y_i = \frac{-1}{n\alpha(n)} \cdot \frac{1-|x|^2}{|x-y|^n} \cdot y_i$$

And finally, we can now compute that $\frac{\partial G}{\partial v}(x,y) = \frac{-1}{n\alpha(n)} \cdot \frac{1-|x|^2}{|x-y|^n} \cdot |y|^2 = \frac{-1}{n\alpha(n)} \cdot \frac{1-|x|^2}{|x-y|^n}$.

With that, we can now conclude the following corollary:

Theorem: Suppose $u \in C^2(\overline{B_1(0)})$ satisfies that $\triangle u = 0$ on $B_1(0)$ and $u|_{\partial B_1(0)} = g$. Then for all $x \in B_1(0)$ we have that:

$$u(x) = \frac{(1-|x|^2)}{n\alpha(n)} \int_{\partial B_1(0)} \frac{g(y)}{|x-y|^n} \mathrm{d}S(y)$$

I need to move onto other topics. So, its time to talk about the heat equation.

All exercises for this class are from the Evan's PDE book (see *citation 16*). Also as a reminder, $\triangle$ and ∇ and $\partial/\partial v$ are always assumed to be derivative operators on only the spatial variables.

Exercise 2.10: Suppose u is smooth and satisfies $u_t - \triangle u = 0$ on $\mathbb{R}^n \times \mathbb{R}_{>0}$.

(a) Show that $w(x,t) := u(\lambda x, \lambda^2 t)$ also satisfies that $w_t - \triangle w = 0$ on $\mathbb{R}^n \times \mathbb{R}_{>0}$ for every $\lambda \in \mathbb{R} - \{0\}$.

Proof:
So long as $\lambda \neq 0$ we know that $\lambda^2 t \in \mathbb{R}_{>0}$ for all $t \in \mathbb{R}_{>0}$. Hence, w is well-defined for all $\lambda \in \mathbb{R} - \{0\}$.

Next, note for any $1 \leq i \leq n$ that $\frac{\partial w}{\partial x_i}(x,t) = \lambda \frac{\partial u}{\partial x_i}(\lambda x, \lambda^2 t)$. In turn:

$$\frac{\partial^2 w}{\partial x_i^2}(x,t) = \lambda^2 \frac{\partial^2 u}{\partial^2 x_i}(\lambda x, \lambda^2 t) \text{ and } \triangle w(x,t) = \lambda^2 \triangle u(\lambda x, \lambda^2 t).$$

Similarly, we get that $w_t(x,t) = \lambda^2 u_t(\lambda x, \lambda^2 t)$. And now we can compute that:

$$w_t(x,t) - \triangle w(x,t) = \lambda^2 (u_t(\lambda x, \lambda^2 t) - \triangle u(\lambda x, \lambda^2 t)) = \lambda^2 \cdot 0 = 0.$$

(b) Show that $w(x,t) := x \cdot \nabla u(x,t) + 2t u_t(x,t)$ satisfies that $w_t - \triangle w = 0$ on $\mathbb{R}^n \times \mathbb{R}_{>0}$.

Proof:
Note that due to the product rule plus the fact that derivatives commute with each other, we know for any $1 \leq i \leq n$ that:

$$\begin{aligned} \partial_{x_i} w(x,t) &= e_i \cdot \nabla u(x,t) + x \cdot \tfrac{\partial}{\partial x_i} \nabla u(x,t) + 2t \partial_{x_i} u_t \\ &= u_{x_i}(x,t) + x \cdot \nabla u_{x_i}(x,t) + 2t (u_{x_i})_t(x,t). \end{aligned}$$

We can similarly take another partial derivative and get that:

$$\begin{aligned} \partial_{x_i}^2 w(x,t) &= u_{x_i,x_i}(x,t) + e_i \cdot \nabla u_{x_i}(x,t) + x \cdot \tfrac{\partial}{\partial x_i} \nabla u_{x_i}(x,t) + 2t \partial_{x_i} (u_{x_i})_t(x,t) \\ &= 2u_{x_i,x_i}(x,t) + x \cdot \nabla u_{x_i,x_i}(x,t) + 2t (u_{x_i,x_i})_t(x,t). \end{aligned}$$

And now we can conclude that:

$$\triangle w(x,t) = 2 \triangle u(x,t) + x \cdot \nabla \triangle u(x,t) + 2t \tfrac{\partial}{\partial t} \triangle u(x,t)$$

Similarly, we can calculate that:

$$w_t(x,t) = x \cdot \nabla u_t(x,t) + 2u_t(x,t) + 2t \tfrac{\partial}{\partial t} u_t(x,t)$$

And now we get that:

$$\begin{aligned} &w_t(x,t) - \triangle w(x,t) \\ &\quad = 2(u_t(x,t) - \triangle u(x,t)) + x \cdot \nabla (u_t - \triangle u)(x,t) + 2t \tfrac{\partial}{\partial t}(u_t - \triangle u)(x,t) \\ &\quad = 2(0) + x \cdot 0 + 2t(0) = 0. \blacksquare \end{aligned}$$

Now recall from *page 921* what the fundamental solution to the equation $u_t - \triangle u = 0$ on $\mathbb{R}^n \times \mathbb{R}_{>0}$ is. In math 148 we denoted it $S^{(n)}$. In math 231 we will denote it Φ.

The last thing I will probably get to take notes on before returning to differential geometry is finally addressing the incomplete proof on *pages 924-925*. For the sake of completeness I will not assume a normalized constant of heat transport when I'm showing the following.

Proposition: Let $\Phi(x,t) = \frac{1}{(2\pi\kappa t)^{n/2}} e^{\frac{-|x|^2}{4\kappa t}}$. Also suppose $f \in C^{2,1}(\mathbb{R}^n \times \mathbb{R}_{\geq 0})$ with f and all its derivatives bounded by a constant. Then:

$$u(x,t) \coloneqq \int_0^t \int_{\mathbb{R}^n} f(y,s)\Phi(x-y,t-s)\mathrm{d}y\mathrm{d}s$$

satisfies that:

$$\lim_{\substack{t\to 0 \\ x'\to x}} u(x',t) = 0, \text{ that } u \in C^{2,1}(\mathbb{R}^n \times \mathbb{R}_{>0}), \text{ and } u_t - \kappa\triangle u = f.$$

Proof:
See the work on *pages 924-925* for why $\lim_{\substack{t\to 0 \\ x'\to x}} u(x',t) = 0$.

Now we just need to show the other claims. To do this, I will first change variables and write $u(x,t) = \int_0^t \int_{\mathbb{R}^n} f(x-y,t-s)\Phi(y,s)\mathrm{d}y\mathrm{d}s$.

We'll in particular write $G(x,t,s) \coloneqq \int_{\mathbb{R}^n} f(x-y,t-s)\Phi(y,s)\mathrm{d}y$.

Note that G_t, G_{x_i}, and G_{x_i,x_j} all exist, equal what they should equal, and are continuous on $\{(x,t,s) : x \in \mathbb{R}^n, t > 0, \text{ and } 0 < s < t\}$.

I'll only show this for the tth partial derivative. Note that if $0 < s_0 < s < t < s_1$ then:

$|f_t(x-y,t-s)\Phi(y,s)| \leq \|f_t\|_u \frac{1}{\sqrt{4\pi\kappa s_0}} e^{\frac{-|y|^2}{4\kappa s_1}}$ where the latter is integrable.

Thus by Leibniz' rule applied to a properly chosen neighborhood containing any (x,t,s) we are interested in, we can conclude that:

$$G_t(x,t,s) = \int_{\mathbb{R}^n} f_t(x-y,t-s)\Phi(y,s)\mathrm{d}y.$$

Also as $f_t(x-y,t-s)\Phi(y,s)$ is continuous, the same bound from before shows that G_t is continuous.

I'll furthermore point out that f can be uniquely extended continuously to $\mathbb{R}^n \times \mathbb{R}_{\geq 0}$ via a result I am about to prove in a tangent, and then nonuniquely extended to a continuous bounded function on all of $\mathbb{R}^n \times \mathbb{R}$ via the Tietz extension theorem. We shall choose such an extension and it doesn't matter which we pick since for the computations of u_t and $\triangle u$ we will only access the value of f on $\mathbb{R}^n \times \mathbb{R}_{\geq 0}$.

I'll actually prove a slightly stronger statement in this tangent because why not?

Proposition: Suppose $D \subseteq \mathbb{R}^n$ is an embedded C^1 submanifold of codimension 0 with a nonempty boundary. If $f : D^\circ \to \mathbb{R}$ is a C^1 map with $|\nabla f|$ bounded on D°, then for all $x \in \partial D$ we have that $\lim_{x'\to x} f(x')$ exists.

Proof:
Let (V,φ) be a smooth chart on D containing x. Then let $\widehat{V} = \varphi(V)$ and $\widehat{x} = \varphi(x)$. Note

that (V, φ) must be a boundary chart as $x \in \partial D$. Hence, without loss of generality we can assume that $\widehat{x} \in \partial \mathbb{R}^n_{\geq 0}$ and $f \circ \varphi^{-1}$ is defined on $\widehat{V} \cap (\mathbb{R}^n_{>0})$. Furthermore, because φ is a homeomorphism, we know that $\lim_{x' \to x} f(x')$ exists iff $\lim_{\widehat{x'} \to \widehat{x}} f \circ \varphi^{-1}(\widehat{x'})$ exists. And finally, if we restrict V to the interior of a compact subneighborhood of x, we can guarentee that $\|Df\varphi^{-1}\|_{\text{op}}$ is bounded on $\widehat{V}$. In turn, we will get that $|\nabla(f \circ \varphi^{-1})|$ is also bounded on D°.

Based on the prior reasoning, it thus suffices to assume D is an open subset of $[0, \infty)^n$ and $f \in C^1(D \cap (0, \infty)^n, \mathbb{R})$ with $|\nabla f|$ bounded by some constant. Also, upon writing $x = (x_1, \ldots, x_n)$, we know at least one x_i is zero. By permuting our coordinates if necessary we can always make it so that specifically $x_n = 0$.

Pick $R > 0$ such that $B_R(x) \cap [0, \infty)^n \subseteq D$. Then make the key observation that $B_R(x) \cap (0, \infty)^n$ is convex. As a result, if $|\nabla f(y)| \leq C$ for all $y \in D$, we have that:

$$|f(y_1) - f(y_2)| \leq C|y_2 - y_1| \text{ for all } y_1, y_2 \in B_R(x) \cap (0, \infty)^n.$$

This lets us conclude that if $\{x_k\}_{k \in \mathbb{N}}$ is *any* sequence in D converging to x, then $\{f(x_k)\}_{k \in \mathbb{N}}$ is a Cauchy sequence (and thus converges).

> Why?
> For any $\varepsilon > 0$ there exists N with $|x_m - x_n| < \varepsilon/C$ for all $m, n \geq N$. In turn, $|f(x_m) - f(x_n)| < \varepsilon$ for all $m, n \geq N$.

Also, if $\{x_k\}_{k \in \mathbb{N}}$ and $\{y_k\}_{k \in \mathbb{N}}$ are differing sequences in D converging to x then $\{f(x_k)\}_{k \in \mathbb{N}}$ and $\{f(y_k)\}_{k \in \mathbb{N}}$ must converge to the same thing.

> Why?
> Suppose to the contrary that they don't and then construct a new sequence by $z_{2k-1} = x_k$ and $z_{2k} = y_k$ for all k. Then $\{f(z_k)\}_{k \in \mathbb{N}}$ will have two distinct subsequential limits (and thus not converge). Yet we also know $\{z_k\}_{k \in \mathbb{N}}$ converges to x in D. This contradicts our prior observation. ■

Continuing where I left off, if $x \in \mathbb{R}^n$, $t \in \mathbb{R}$, and $0 < s_0 < s < s_1$ then:

$$|f(x - y, t - s)\Phi(y, s)| \leq \|f\|_u \frac{1}{\sqrt{4\pi\kappa s_0}} e^{\frac{-|y|^2}{4\kappa s_1}}$$

So by Leibniz' rule, we know that G is a well-defined continuous function on the set $\{(x, t, s) \in \mathbb{R}^n \times \mathbb{R} \times (0, \infty)\}$.

Now, we're ready to define $H(x, t, a) \coloneqq \int_0^a G(x, t, s)\mathrm{d}s$ on $\mathbb{R}^n \times \mathbb{R} \times (0, \infty)$. Importantly, $\|G\|_u \leq \|f\|_u$ for all $(x, t, s) \in \mathbb{R}^n \times \mathbb{R} \times (0, \infty)$. So, H is well-defined and continuous. Plus, by the fundamental theorem of calculus we have that:

$$\frac{\partial H}{\partial a}(x, t, a) = G(x, t, a) \text{ for all } (x, t, a) \in \mathbb{R}^n \times \mathbb{R} \times (0, \infty).$$

Next note that $|G_t(x, t, s)| \leq \|f_t\|_u$ for all:

$$(x, t, s) \in \{(x, t, s) : x \in \mathbb{R}^n, t > 0, \text{ and } 0 < s < t\}.$$

This is enough to show by Leibniz's rule that for any $x \in \mathbb{R}^n$, $t > 0$, and $0 < a \leq t$ that $\frac{\partial H}{\partial t}(x, t, a)$ exists and equals $\int_0^a G_t(x, t, s)\mathrm{d}s$.

Now, we can apply chain rule to get that if $x \in \mathbb{R}^n$ and $t > 0$ then:

$$u_t(x,t) = \frac{\partial H(x,t,t)}{\partial t} = \int_0^t G_t(x,t,s)\mathrm{d}s + G(x,t,t)$$
$$= \int_0^t \int_{\mathbb{R}^n} f_t(x-y,t-s)\Phi(y,s)\mathrm{d}y\mathrm{d}s + \int_{\mathbb{R}^n} f(x-y,0)\Phi(y,t)$$

Meanwhile, because $|G_{x_i}(x,t,s) \leq \|f_{x_i}\|_u$ and $|G_{x_i,x_j}(x,t,s)| \leq \|f_{x_i,x_j}\|_u$ for all:

$$(x,t,s) \in \{(x,t,s) : x \in \mathbb{R}^n, t > 0, \text{ and } 0 < s < t\},$$

we can analogously apply Leibniz's rule to get that:

$$u_{x_i,x_i}(x,t) = \frac{\partial^2 H}{\partial x_i^2}(x,t,t) = \int_0^t G_{x_i,x_i}(x,t,s)\mathrm{d}s$$
$$= \int_0^t \int_{\mathbb{R}^n} f_{x_i,x_i}(x-y,t-s)\Phi(y,s)\mathrm{d}y\mathrm{d}s.$$

And with that the hard part is now done because we have successfully moved the derivative operators inside our integral expression and proven that:

$$(\partial_t - \triangle)u(x,t)$$
$$= \int_0^t \int_{\mathbb{R}^n} (\partial_t - \triangle)f(x-y,t-s)\Phi(y,s)\mathrm{d}y\mathrm{d}s + \int_{\mathbb{R}^n} f(x-y,0)\Phi(y,t)\mathrm{d}y.$$

Now we still need to actually show for any fixed $(x,t) \in \mathbb{R}^n \times \mathbb{R}_{>0}$ that our awful integral expression evaluates to $f(x,t)$. To do this, we consider any fixed $0 < \varepsilon < t$ and then we set:

- $I_\varepsilon := \int_\varepsilon^t \int_{\mathbb{R}^n} (\partial_t - \triangle)f(x-y,t-s)\Phi(y,s)\mathrm{d}y\mathrm{d}s$
- $J_\varepsilon := \int_0^\varepsilon \int_{\mathbb{R}^n} (\partial_t - \triangle)f(x-y,t-s)\Phi(y,s)\mathrm{d}y\mathrm{d}s$
- $K := \int_{\mathbb{R}^n} f(x-y,0)\Phi(y,t)\mathrm{d}y.$

Note that $(\partial_t - \triangle)u(x,t) = I_\varepsilon + J_\varepsilon + K$.

Claim 1: $J_\varepsilon \to 0$ as $\varepsilon \to 0^+$.

Why?

We know from our assumptions on f that $\|f_t - \triangle f\|_u < \infty$. In turn, we have that:

$$|J_\varepsilon| \leq \varepsilon \|f_t - \triangle f\|_u \to 0 \text{ as } \varepsilon \to 0$$

Claim 2: $I_\varepsilon \to f(x,t) - K$ as $\varepsilon \to 0^+$.

Why?

Observe that:

$$\int_\varepsilon^t \int_{\mathbb{R}^n} f_t(x-y,t-s)\Phi(y,s)\mathrm{d}y\mathrm{d}s$$
$$= \int_{\mathbb{R}^n} \int_\varepsilon^t f_t(x-y,t-s)\Phi(y,s)\mathrm{d}s\mathrm{d}y \text{ (by Fubini)}$$
$$= -\int_{\mathbb{R}^n} \int_\varepsilon^t \frac{\partial f(x-y,t-s)}{\partial s}\Phi(y,s)\mathrm{d}s\mathrm{d}y$$
$$= -\int_{\mathbb{R}^n} f(x-y,0)\Phi(y,t)\mathrm{d}y + \int_{\mathbb{R}^n} f(x-y,t-\varepsilon)\Phi(y,\varepsilon)\mathrm{d}y$$
$$+ \int_{\mathbb{R}^n} \int_\varepsilon^t f(x-y,t-s)\Phi_t(y,s)\mathrm{d}s\mathrm{d}y$$
(by integration by parts)
$$= -K + \int_{\mathbb{R}^n} f(x-y,t-\varepsilon)\Phi(y,\varepsilon)\mathrm{d}y$$
$$+ \int_\varepsilon^t \int_{\mathbb{R}^n} f(x-y,t-s)\Phi_t(y,s)\mathrm{d}y\mathrm{d}s \text{ (by Fubini).}$$

Meanwhile, because Φ and all its derivatives vanish at infinity, we can conclude via integration by parts that for any $1 \leq i \leq n$ and fixed $\varepsilon \leq s \leq t$ that:

$$\begin{aligned}\int_{\mathbb{R}^n} f_{x_i,x_i}(x-y,t-s)\Phi(y,s)\mathrm{d}y &= -\int_{\mathbb{R}^n} \frac{\partial(f_{x_i}(x-y,t-s))}{\partial y_i}\Phi(y,s)\mathrm{d}y \\ &= \int_{\mathbb{R}^n} f_{x_i}(x-y,t-s)\Phi_{y_i}(y,s)\mathrm{d}y \\ &= -\int_{\mathbb{R}^n} \frac{\partial f(x-y,t-s)}{\partial y_i}\Phi_{y_i}(y,s)\mathrm{d}y \\ &= \int_{\mathbb{R}^n} f(x-y,t-s)\Phi_{y_i,y_i}(y,s)\mathrm{d}y.\end{aligned}$$

So in conclusion, we get that:

$$I_\varepsilon = -K + \int_{\mathbb{R}^n} f(x-y,t-\varepsilon)\Phi(y,\varepsilon)\mathrm{d}y + \int_\varepsilon^t \int_{\mathbb{R}^n} f(x-y,t-s)(\Phi_t - \triangle\Phi)(y,s)\mathrm{d}y\mathrm{d}s$$

But now Φ satisfies the (homogeneous) heat equation. So the last integral cancels and we are left with:

$$I_\varepsilon = \int_{\mathbb{R}^n} f(x-y,t-\varepsilon)\Phi(y,\varepsilon)\mathrm{d}y - K.$$

Now note that:

$$\begin{aligned}&\int_{\mathbb{R}^n} f(x-y,t-\varepsilon)\Phi(y,\varepsilon)\mathrm{d}y \\ &\qquad = \int_{\mathbb{R}^n} \left[f(x-y,t-\varepsilon) - f(x-y,t)\right]\Phi(y,\varepsilon)\mathrm{d}y + \int_{\mathbb{R}^n} f(x-y,t)\Phi(y,\varepsilon)\mathrm{d}y\end{aligned}$$

I claim that $\int_{\mathbb{R}^n} \left[f(x-y,t-\varepsilon) - f(x-y,t)\right]\Phi(y,\varepsilon)\mathrm{d}y \to 0$ as $\varepsilon \to 0^+$.

> After all, since f is continuous (and thus in particular uniformly continuous on a compact ball containing (x,t)) we can for any $\kappa > 0$ pick some $\delta > 0$ such that $|f(x-y,t-\varepsilon) - f(x-y,t)| < \kappa$ for all $y \in \overline{B_\delta}(x)$ and $\varepsilon < \delta$. In turn for $\varepsilon < \delta$ we have that:
>
> $$\begin{aligned}&\left|\int_{\overline{B_\delta(x)}} \left[f(x-y,t-\varepsilon) - f(x-y,t)\right]\Phi(y,\varepsilon)\mathrm{d}y\right| \\ &\qquad \leq \kappa \int_{\overline{B_\delta(x)}} \Phi(y,\varepsilon)\mathrm{d}y \leq \kappa \int_{\mathbb{R}^n} \Phi(y,\varepsilon)\mathrm{d}y = \kappa.\end{aligned}$$
>
> Yet simultaneously, because $|f(x-y,t-\varepsilon) - f(x-y,t)|$ is bounded by $2\|f\|_u$, it is an easy to establish fact that:
>
> $$\lim_{\varepsilon\to 0^+} \int_{B_\delta(x)^c} \left[f(x-y,t-\varepsilon) - f(x-y,t)\right]\Phi(y,\varepsilon)\mathrm{d}y \to 0 \text{ as } \varepsilon \to 0$$
>
> Therefore:
>
> $$\begin{aligned}&\limsup_{\varepsilon\to 0^+} \left|\int_{\mathbb{R}^n} \left[f(x-y,t-\varepsilon) - f(x-y,t)\right]\Phi(y,\varepsilon)\mathrm{d}y\right| \\ &\qquad = \limsup_{\varepsilon\to 0^+} \left|\int_{\overline{B_\delta(x)}} \left[f(x-y,t-\varepsilon) - f(x-y,t)\right]\Phi(y,\varepsilon)\mathrm{d}y\right| \leq \kappa\end{aligned}$$
>
> And since κ was arbitrary, we can take $\kappa \to 0$ to finish showing this claim.

Meanwhile, note that $w(x,\varepsilon) := \int_{\mathbb{R}^n} f(x-y,t)\Phi(y,\varepsilon)\mathrm{d}y$ is just the solution to the (homogeneous) heat equation $w_t - \triangle w = 0$ with an initial value of $w(x,0) = f(x,t)$. So we can already identify from the proof on *pages 921-922* that $\int_{\mathbb{R}^n} f(x-y,t)\Phi(y,\varepsilon)\mathrm{d}y \to f(x,t)$ as $\varepsilon \to 0^+$.

This finishes showing that $I_\varepsilon \to f(x,t) - K$ as $\varepsilon \to 0$.

And now adding together the limits we just proved finishes showing that:

$$u_t(x,t) - \triangle u(x,t) = f(x,t). \blacksquare$$

10/23/2026

Math 250A Lecture Notes:
Today and maybe tomorrow I need to take lecture notes for this class and also do homework. Once I've dealt with that, I can go back to taking PDE notes.

To start off, I want to give an actual atlas for $S^n \subseteq \mathbb{R}^{n+1}$. Note that this is in fact an embedded submanifold as it is the regular zero set of $\Phi : \mathbb{R}^{n+1} \to \mathbb{R}$ given by $\Phi(x) = |x|^2 - 1$. Note that all the following reasoning is heavily inspired by work on *page 241* showing that $\mathbb{C}_\infty$ is homeomorphic to S^2.

Let $U_1 = S^n - \{(0, \dots, 0, 1)\}$ and let $\widehat{U}_1 = \mathbb{R}^n$. We define:

- $\varphi_1 : \mathbb{R}^{n+1} - \{(0, \dots, 0, 1)\} \to \widehat{U}_1$ by:
$$\varphi_1(x_1, \dots, x_n, x_{n+1}) := \left(\frac{x_1}{1-x_{n+1}}, \dots, \frac{x_n}{1-x_{n+1}}\right)$$
- $\psi_1 : \widehat{U}_1 \to \mathbb{R}^{n+1}$ by $\psi_1(x_1, \dots, x_n) := \left(\frac{2x_1}{1+|x|^2}, \dots, \frac{2x_n}{1+|x|^2}, \frac{|x|^2-1}{1+|x|^2}\right)$.

Note that φ_1 is a smooth map. It follows that it's restriction to U_1 will still be a smooth map with respect to the smooth structure on the open submanifold $U_1 \subseteq S^n \subseteq \mathbb{R}^{n+1}$.

Meanwhile, note that ψ_1 is a smooth map. Furthermore:
$$|\psi_1(x)|^2 = \frac{1}{(1+|x|^2)^2}(4|x|^2 + (|x|^2 - 1)^2) = \frac{1}{(1+|x|^2)^2}(|x|^4 + 2|x|^2 + 1) = 1.$$

So, ψ_1 maps into S^n.

We can further note that $(0, \dots, 0, 1) \notin \psi_1(\widehat{U}_1)$. And since $U_1 \subseteq S^n \subseteq \mathbb{R}^{n+1}$ is an *embedded* submanifold, we can thus conclude that restricting the codomain of ψ_1 does not change that ψ_1 is a smooth map.

With that, we've proven that $\varphi_1 : U_1 \to \widehat{U}_1$ and $\psi_1 : \widehat{U}_1 \to U_1$ are both smooth maps. Finally, I claim that $\psi_1 \circ \varphi_1 = \mathrm{Id}_{U_1}$ and $\varphi_1 \circ \psi_1 = \mathrm{Id}_{\widehat{U}_1}$.

You do this computation yourself. There's no fancy ideas.

And with that, we've proven that (U_1, φ_1) is a smooth chart on S^n.

In a similar vein, if $U_2 = S^n - \{(0, \dots, 0, -1)\}$ then we can define another smooth chart (U_2, φ_2) on S^n by:

- $\varphi_2(x_1, \dots, x_{n+1}) = \left(\frac{x_1}{1+x_{n+1}}, \dots, \frac{x_n}{1+x_{n+1}}\right)$,
- $\varphi_2^{-1}(x_1, \dots, x_n) = \left(\frac{2x_1}{1+|x|^2}, \dots, \frac{2x_n}{1+|x|^2}, \frac{1-|x|^2}{1+|x|^2}\right)$.

The prior two charts (U_1, φ_1) and (U_2, φ_2) are called the stereographic projections and they let you cover S^n with the fewest possible charts.

(We can't cover S^n with just a single chart (smooth or not) because that would imply the compact set S^n is homeomorphic to an open subset of $\mathbb{R}^n$ (since S^n has no boundary). But that's impossible.)

Sometimes, we want to use a smooth graph chart instead though. I'll construct that here.

Let $U = \{(x_1, \ldots, x_{n+1}) \in S^n : x_{n+1} > 0\}$. Next set $\widehat{U} = B_1(0) \subseteq \mathbb{R}^n$. And finally, define the maps:

- $\varphi : \mathbb{R}^{n+1} \to \mathbb{R}^n$ by $\varphi(x_1, \ldots, x_n, x_{n+1}) := (x_1, \ldots, x_n)$,
- $\psi : \widehat{U} \to \mathbb{R}^{n+1}$ by $\psi(x_1, \ldots, x_n) = (x_1, \ldots, x_n, \sqrt{1 - x_1^2 - \ldots - x_n^2})$.

By identical reasoning as before, these become smooth maps from U to $\widehat{U}$ and $\widehat{U} \to U$ respectively. And identically to before, we can prove that $\psi \circ \varphi = \mathrm{Id}_U$ and $\varphi \circ \psi = \mathrm{Id}_{\widehat{U}}$.

This finishes showing that $(U, \varphi|_U)$ is a smooth chart on S^n.

In the prior construction, we can also obviously make the dependent variable in our graph any x_i instead of just x_{n+1}. Also, instead of considering the open set $\{(x_1, \ldots, x_{n+1}) \in S^n : x_{n+1} > 0\}$ we can insetad consider the open set $\{(x_1, \ldots, x_{n+1}) \in S^n : x_{n+1} < 0\}$. And then we just have to negate $\sqrt{1 - x_1^2 - \ldots - x_n^2}$ in our expression for the inverse map.

Set 3 Problem 1: Note that Oprea writes $\mathrm{span}((x_1, \ldots, x_{n+1})) \in \mathbb{RP}^n$ as $[x_1 : \cdots : x_{n+1}]$.

(i) Show that the map $q : S^2 \to \mathbb{RP}^2$ given by $q(x, y, z) = [x : y : z]$ is a smooth local diffeomorphism.

Given any $(x_0, y_0, z_0) \in S^2$ we know at least one coordinate is nonzero. Without loss of generality, I will assume $z_0 > 0$. After all, the argument remains basically identical if $z_0 < 0$. And if $z_0 = 0$ then we can just rename our coordinates and do the exact same proof on our renamed coordinates.

Let $U = \{(x, y, z) \in S^2 : z > 0\}$. Then we have a smooth parametrization $X : B_1(0) \subseteq \mathbb{R}^2 \to U$ given by $X(x, y) = (x, y, \sqrt{1 - x^2 - y^2})$. In other words, (U, X^{-1}) is a smooth chart on S^2 containing (x_0, y_0, z_0).

Meanwhile, recall that $V = \{[x : y : z] \in \mathbb{RP}^2 : z \neq 0\}$ and $\psi : V \to \mathbb{R}^2$ given by $\psi([x : y : z]) = (\frac{x}{z}, \frac{y}{z})$ define a smooth chart on $\mathbb{RP}^2$. In particular, we have that $q(U) \subseteq V$.

To check that q is smooth and a local diffeomorphism, it suffices to show that q is smooth with respect to the local coordinates (U, X^{-1}) and (V, ψ), and also that $\mathrm{D}(\psi \circ q \circ X)$ is invertible on $B_1(0)$. Fortunately, this is easy.

Note that $\widehat{q} := \psi \circ q \circ X : B_1(0) \to \mathbb{R}^2$ is given by:

$$\widehat{q}(x, y) = (\tfrac{x}{\sqrt{1-x^2-y^2}}, \tfrac{y}{\sqrt{1-x^2-y^2}}).$$

This is a smooth map on $B_1(0)$. Also:

$$\mathrm{D}\widehat{q}(x, y) = \begin{bmatrix} \frac{1-y^2}{(1-x^2-y^2)^{3/2}} & \frac{xy}{(1-x^2-y^2)^{3/2}} \\ \frac{xy}{(1-x^2-y^2)^{3/2}} & \frac{1-x^2}{(1-x^2-y^2)^{3/2}} \end{bmatrix} = \frac{1}{(1 - x^2 - y^2)^{3/2}} \begin{bmatrix} 1 - y^2 & xy \\ xy & 1 - x^2 \end{bmatrix}$$

Since $\frac{1-y^2}{(1-x^2-y^2)^{3/2}} \neq 0$ ever, we can thus conclude that $\mathrm{D}\widehat{q}(x,y)$ is singular iff:

$$\begin{bmatrix} 1-y^2 & xy \\ xy & 1-x^2 \end{bmatrix} \text{ is singular.}$$

This happens iff $0 = (1-y^2)(1-x^2) - x^2y^2 = 1 - x^2 - y^2$. Yet the latter never happens for all $(x,y) \in B_1(0)$.

(ii) Let $f : S^2 \to \mathbb{R}^4$ be given by $f(x,y,z) = (x^2-y^2, xy, xz, yz)$. Show that f is a smooth immersion.

To start off, let $\iota : S^2 \hookrightarrow \mathbb{R}^3$ be the embedding map. Also let $F : \mathbb{R}^3 \to \mathbb{R}^4$ be defined by $F(x,y,z) = (x^2-y^2, xy, xz, yz)$. Then we have that $f = F \circ \iota$. Note that because F an ι are both smooth, we trivially have that f is smooth. The harder part is showing that f is an immersion.

Now $\mathrm{D}F(x,y,z) = \begin{bmatrix} 2x & -2y & 0 \\ y & x & 0 \\ z & 0 & x \\ 0 & z & y \end{bmatrix}$.

Based on the above expression, we can now conclude that $\mathrm{D}F$ is full rank at most points of S^2. After all:

- If $x \neq 0$, $y \neq 0$, and $z \neq 0$, then the determinant of the bottom three rows is $-2xyz \neq 0$.
- If $z = 0$ then our matrix simplifies and we can show that $\mathrm{D}F(x,y,z)$ has full rank if and only if either $x \neq 0$ or $y \neq 0$.
- If $x = 0$ and $y \neq 0$ and $z \neq 0$ then we can also simplify our matrix and show that $\mathrm{D}F(x,y,z)$ has full rank.
- And if $y = 0$ and $x \neq 0$ and $z \neq 0$, then we similarly can show that $\mathrm{D}F(x,y,z)$ has full rank.

In other words, the only points of S^2 where $\mathrm{D}F$ is not full rank are $(0,0,1)$ and $(0,0,-1)$. Since a composition of injective linear maps is injective, we can thus conclude that $\mathrm{d}f_p$ is injective for all $p \in S^2 - \{(0,0,1),(0,0,-1)\}$.

As for showing that $\mathrm{d}f_{(0,0,\pm 1)}$ is injective, we need to show that:

$$T_{(0,0,\pm1)}S^2 \cap \ker(\mathrm{d}F_{(0,0,\pm1)}) = \{0\}.$$

Fortunately, S^2 is regular zero set of $\Phi : \mathbb{R}^3 \to \mathbb{R}$ given by $\Phi(x) = |x|^2 - 1$. Therefore:

$$T_{(0,0,\pm1)}S^2 = \ker(\mathrm{D}\Phi(0,0,\pm1)) = \ker([0 \quad 0 \quad \pm 2]) = \mathbb{R}^2 \times \{0\}$$

Meanwhile, $\ker(\mathrm{d}F_{(0,0,\pm1)}) = \ker\left(\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ \pm 1 & 0 & 0 \\ 0 & \pm 1 & 0 \end{bmatrix}\right) = \{(0,0)\} \times \mathbb{R}$.

(c) Deduce there exists an embedding $g : \mathbb{RP}^2 \to \mathbb{R}^4$ such that $g \circ q = f$.

From part (a) we know that q is a smooth submersion from S^2 to $\mathbb{RP}^2$. Its pretty trivial that q is also surjective. Thirdly, note that the fibers of q are precisely the sets $\{\vec{x}, -\vec{x}\}$ for any $\vec{x} \in S^2$. This is notable because we in turn can see that f is constant on each of the fibers of q.

By corollary 4.30 on *pages 1108-1109*, we thus know there is a unique smooth map $g : \mathbb{RP}^2 \to \mathbb{R}^4$ satisfying that $g \circ q = f$.

Now to show that g is an immersion, note for any $[x] \in \mathbb{RP}^2$ that we can pick $x \in S^2$ with $q(x) = [x]$. Then $\mathrm{d}g_{[x]} \circ \mathrm{d}q_x = \mathrm{d}f_x$. Yet because q is a local diffeomorphism, we know $\mathrm{d}q_x$ is invertible. So, $\mathrm{d}g_{[x]} = \mathrm{d}f_x \circ (\mathrm{d}q_x)^{-1}$ is the composition of two injective linear maps and is thus injective.

Next, to prove that g is an injection it suffices to show that if $f(x_1, y_1, z_1) = f(x_2, y_2, z_2)$ then (x_1, y_1, z_1) and (x_2, y_2, z_2) are in the same fiber of p.

In other words, we need to show that $(x_2, y_2, z_2) = \pm(x_1, y_1, z_1)$.

To do this, recall from math 220 that the map $s(x, y) \coloneqq (x^2 - y^2, 2xy)$ acts homeomorphically (and in fact biholomorphically) from:

- $\mathbb{R} \times (0, \infty) \to \mathbb{R}^2 - \{(x, 0) : x \geq 0\}$,
- $\mathbb{R} \times (-\infty, 0) \to \mathbb{R}^2 - \{(x, 0) : x \geq 0\}$.

Scaling the y axis afterwards doesn't modify the codomain. So, the fiber of any $(x', y') \in \mathbb{R}^2 - \{(x, 0) : x \geq 0\}$ under intersects $\mathbb{R}^2 - \{\mathbb{R} \times \{0\}\}$ at exactly two points.

Meanwhile, if $y = 0$ then we know $(x^2 - y^2, xy) = (x^2, 0) \in (0, \infty) \times \{0\}$. So, none of the $(x, 0)$ contribute any points to the fibers in the prior paragraph. Also, the fiber of every $(x', 0) \in \mathbb{R}^2 - \{(x, 0) : x > 0\}$ has exactly two points. Meanwhile, the fiber of $(0, 0)$ is just $\{(0, 0)\}$.

With that, we've shown that if $h : \mathbb{R}^2 \to \mathbb{R}^2$ is given by $h(x, y) = (x^2 - y^2, xy)$ then $h^{-1}(\{(x', y')\})$ has exactly two points unless $(x', y') = (0, 0)$.

Finally, note that $h(x, y) = h(-x, -y)$ for all $(x, y) \in \mathbb{R}^2$. Hence, we've proven that if $h(x_1, y_1) = a = h(x_2, y_2)$ then $(x_1, y_1) = \pm(x_2, y_2)$.

Now the prior reasoning was relevant because if $f(x_1, y_1, z_1) = f(x_2, y_2, z_2)$ then automatically $h(x_1, y_1) = h(x_2, y_2)$. So, there exists $\kappa \in \{-1, +1\}$ with $x_2 = \kappa x_1$ and $y_2 = \kappa y_1$.

Now suppose $x_1 \neq 0$. Then $x_1 z_1 = x_2 z_2 = \kappa x_1 z_2 \implies z_1 = \kappa z_2$. But $\kappa = \kappa^{-1}$. So, we can conclude $z_2 = \kappa z_1$ as well.

Similarly, if $y_1 \neq 0$ then we can conclude that $z_2 = \kappa z_1$.

Finally, suppose that $x_1 = y_1 = 0$. Then we know $z_1, z_2 \in \{-1, +1\}$ since otherwise $(x_i, y_i, z_i) \notin S^2$. And thus, the claim becomes trivial that $(x_1, y_1, z_1) = \pm(x_2, y_2, z_2)$.

Finally, since g is a smooth injective immersion from the manifold $\mathbb{RP}^2$ which happens to be compact, we can conclude by *problem 5-3* that g is a smooth embedding. ■

The next exercise will be important to proving a theorem in class.

Set 3 Problem 5: Let M be a C^r manifold (where $r \geq 1$) and let $\delta : M \to (0, \infty)$ be a continuous function. Suppose $f : M \to \mathbb{R}^k$ is another continuous function. Then there is a C^r map $F : M \to \mathbb{R}^k$ such that $|F(x) - f(x)| < \delta(x)$ for all $x \in M$.

Proof:

Firstly, I shall find a sequence of points $\{a_n\}_{n\in\mathbb{N}}$ and open sets $\{U_n\}_{n\in\mathbb{N}}$ such that: $\bigcup_{n\in\mathbb{N}} U_n = M$, $a_n \in U_n$ for all n, and $|f(x) - f(a_n)| < \delta(x)$ for all $x \in U_n$.

Note that if $a \in M$ then we know that $\delta(a) > 0$. And now as δ is continuous, for any $0 < r < \delta(a)$ we can find an open set V_a containing a with $\delta(x) > r$ for all $x \in V_a$. Next, because f is continuous at a, we can find an open set $U_a \subseteq V_a$ containing a such that $|f(x) - f(a)| < r$ for all $x \in U_a$. In turn, $|f(x) - f(a)| < \delta(x)$ for all $x \in U_a$.

The prior paragraph shows us how to give an open covering of M consisting of open sets $\{U_a\}_{a\in M}$ satisfying that $a \in U_a$ for all a and $|f(x) - f(a)| < \delta(x)$ for all $x \in U_a$. Now we just use the second countability of M in order to take a countable subcover.

Now that I have the prior sequence, I'm going to construct a C^r partition of unity $\{\phi_n\}_{n\in\mathbb{N}}$ subordinate to the open sets U_n. Then, I'm going to define $F : M \to \mathbb{R}^k$ by:

$$F(x) = \sum_{n=0}^{\infty} \phi_n(x) f(a_n).$$

Note that since any $x \in M$ has a neighborhood on which only finitely many ϕ_n are nonzero, we can say that locally F is a linear combination of C^r maps. Hence, F is also a C^r map.

Also for any fixed $x \in X$, let $n_1, \ldots, n_k$ be the finitely many integers such that $\phi_{n_i}(x) \neq 0$. Then we have that:

$$\begin{aligned} |F(x) - f(x)| &= \left|\sum_{i=1}^{k} \phi_{n_i}(x) \cdot [f(a_n) - f(x)]\right| \\ &\leq \sum_{i=1}^{k} \phi_{n_i}(x)|f(x) - f(a_n)| < \sum_{i=1}^{k} \phi_{n_i}(x)\delta(x) = \delta(x). \blacksquare \end{aligned}$$

The final exercises on the homework all are about introducing Lie groups. Since 251 and not 250 is the Lie groups class, I don't think these will come up again in my differential geometry class. I'll still take notes on this though.

Let G be a group (written multiplicatively). We let $m : G \times G \to G$ denote the multiplication map and $i : G \to G$ denote the inverse map. Also let e denote the identity of G. If G is also a smooth manifold and m and i are C^∞ functions, we say G is a Lie group.

Note in particular that all Lie groups are topological groups.

A special property of a Lie group G is that the group of smooth diffeo-automorphisms on G acts transitively on G. After all, given any $g \in G$ we define $L_g : G \to G$ and $R_g : G \to G$ by:

$$L_g(h) \coloneqq gh \text{ and } R_g(h) \coloneqq hg.$$

Note that L_g is the composition of the smooth map $h \mapsto (g, h)$ and the smooth map m. So, L_g is a smooth map. We similarly can show that R_g is a smooth map. Also, both L_g and R_g are diffeomorphisms since they have smooth inverses given by $L_{g^{-1}}$ and $R_{g^{-1}}$.

Now if we consider the obvious action of diffeo-automorphisms of G on G, we can see for any $g \in G$ that $L_g(e) = g$ for all g. This proves that the orbit of e is all of G.

Corollary: Lie groups have no boundary.

Proof:
Let g be an interior point of G and then suppose h is any other point in G. We know there is a diffeomorphism $F : G \to G$ satisfying that $f(h) = g$. Yet diffeomorphisms must map interior points to interior points (see exercise 4-2 on *page 1057*). So, we know h is an interior point of G. ■

A homomorphism $f : G \to H$ of Lie groups is a group homomorphism that is also a smooth map. An isomorphism of Lie groups is similarly a group isomorphism which is also a smooth diffeomorphism.

A Lie subgroup H of G is a subgroup of G endowed with a smooth structure making H into a Lie group and an immersed submanifold of G.

Set 3 Problem 2: Let G be a Lie group.

(i) Show that if H is a weakly embedded smooth submanifold of G that is also a subgroup of G, then H is a Lie subgroup of G.

Lemma: Suppose M_1, M_2, N_1, N_2 are C^r manifolds. If $F : M_1 \to N_1$ and $G : M_2 \to N_2$ are both C^r immersions, then $F \times G : M_1 \times M_2 \to N_1 \times N_2$ is also a C^r immersion.

Proof:
Let $(p, q) \in M_1 \times M_2$ and recall that there are standard identifications:

- $T_{(p,q)}(M_1 \times M_2) \cong T_pM_1 \oplus T_qM_2$,
- $T_{(F(p),G(q))}(N_1 \times N_2) \cong T_{F(p)}N_1 \oplus T_{G(q)}N_2$.

For each i let $\pi_i : M_1 \times M_2 \to M_i$ and $\widetilde{\pi}_i : N_1 \times N_2 \to N_i$ be the projection maps. Then importantly we have that $F \circ \pi_1 = \widetilde{\pi}_1 \circ (F \times G)$ and $G \circ \pi_2 = \widetilde{\pi}_2 \circ (F \times G)$. As a result for any $v \in T_{(p,q)}(M_1 \times M_2)$ we have that:

$$\begin{aligned}
\mathrm{d}(F \times G)_{(p,q)}(v) &= (\mathrm{d}(\widetilde{\pi}_1)_{(F(p),G(p))} \circ \mathrm{d}(F \times G)_{(p,q)}(v),\ \ \mathrm{d}(\widetilde{\pi}_2)_{(F(p),G(p))} \circ \mathrm{d}(F \times G)_{(p,q)}(v)) \\
&= (\mathrm{d}(\widetilde{\pi}_1 \circ (F \times G))_{(p,q)}(v),\ \ \mathrm{d}(\widetilde{\pi}_2 \circ (F \times G))_{(p,q)}(v)) \\
&= (\mathrm{d}(F \circ \pi_1)_{(p,q)}(v),\ \ \mathrm{d}(G \circ \pi_2)_{(p,q)}(v)) \\
&= (\mathrm{d}F_p \circ \mathrm{d}(\pi_1)_{(p,q)}(v),\ \ \mathrm{d}G_q \circ \mathrm{d}(\pi_2)_{(p,q)}(v)).
\end{aligned}$$

In other words, according to the standard identifications we have that:

$$\mathrm{d}(F \times G)_{(p,q)}(v_1, v_2) = (\mathrm{d}F_p(v_1), \mathrm{d}G_q(v_2)).$$

Even more succinctly, this means $\mathrm{d}(F \times G)_{(p,q)} = \mathrm{d}F_p \times \mathrm{d}G_q$ under this identication. Finally, note that:

$$\ker(\mathrm{d}F_p \times \mathrm{d}G_q) = \ker(\mathrm{d}F_p) \oplus \ker(\mathrm{d}G_q) = \{0\} \oplus \{0\} = \{0\}.$$

So, we've finished showing that $\mathrm{d}(F \times G)_{(p,q)}$ is injective.

I went to the effort of showing the prior lemma because we can now use it to conclude that $H \times H$ is an immersed submanifold of $G \times G$. In turn, if $m : G \times G \to G$ is a smooth map then so is $m|_{H \times H} : H \times H \to G$.

Next, as H is a subgroup of G, we know that the image of $m|_{H \times H}$ is contained in H. Therefore, as H is weakly embedded we know that we can restrict the codomain of $m|_{H \times H}$ in order to get that $m|_{H \times H} : H \times H \to H$ is a smooth map.

We can use analogous reasoning to show that the inversion map i is smooth from H into H.

(b) Give an example of a dense immersed Lie subgroup of the torus $\mathbb{T} := S^1 \times S^1$.

Let $\alpha \in \mathbb{R} - \mathbb{Q}$ and then define $\gamma : \mathbb{R} \to \mathbb{T}$ by $\gamma(t) = (e^{2\pi i t}, e^{2\pi i \alpha t})$. We know from set 2 problem 7 on *pages 1214-1216* that γ is an injective immersion whose image is dense in $\mathbb{T}$. Therefore, $H := \gamma(\mathbb{R})$ is an immersed submanifold of $\mathbb{T}$ that is dense in $S^1 \times S^1$.

Next, note that $\mathbb{T}$ is a Lie group under pointwise complex multiplication.

I don't feel like proving it since it seems obvious to me.

Furthermore, note that $\mathbb{R}$ is a Lie group under addition. Plus, $\gamma : \mathbb{R} \to \mathbb{T}$ is a group homomorphism. Thus, H is automatically a subgroup of $\mathbb{T}$. And by definition we have that γ is a smooth diffeomorphism from $\mathbb{R}$ to H.

It follows that multiplication and inversion are continuous with respect to the topology on H because addition and negation is smooth with respect to the topology on $\mathbb{R}$.

(c) If G is a Lie group, show that the connected component G^0 of the identity element e is a Lie subgroup. Furthermore, every other connected component of G is diffeomorphic to G^0.

To start off, because G is locally connected (because it is a manifold), we know that every connected component of G is an open set. Thus, G^0 is an open embedded submanifold of G^0.

Next, I claim that if $g \in G^0$ then so is g^{-1}. To see why, note that $i : G \to G$ is in particular a homeomorphism. So, it must permute the connected components of G. Yet also note that $e \in G^0$ and $i(e) = e$. So i must map G^0 back to G^0.

In a similar vein, I claim that if $g, h \in G^0$ then $gh \in G^0$ as well. After all, consider any fixed $g \in G^0$ and let $L_g : G \to G$ be the map $L_g(h) = gh$. Then L_g is in particular a homeomorphism. So, it also must permute the connected components of G. Yet $e \in G^0$ and $L_g(e) = g \in G^0$. So, L_g must map G^0 back to G^0. So, in particular, if $h \in G^0$ then $gh \in G^0$.

With that, we can now conclude by part (a) that G^0 is a Lie subgroup of G.

Finally, suppose G^1 is another connected component of G and let $g' \in G^1$. Then define $L_{g'} : G \to G$ by $L_{g'}(h) = g'h$. Because $L_{g'}$ is a homeomorphism, it must permute the connected components of G. Yet $e \in G^0$ and $L_{g'}(e) = g' \in G^1$. So, $L_{g'}(G^0) = G^1$. And now by restricting the domain and codomain of $L_{g'}$, we get that $L_{g'}$ is a smooth diffeomorphism from G^0 to G^1. ■

Let G be a Lie group and X a C^r manifold. We say G acts on X (on the left) if we have a C^r map $G \times X \to X$ written as $(g, x) \mapsto gx$ satisfying the standard group action axioms.

If X, Y are two C^r manifolds being acted on by a Lie group G and $f : X \to Y$ is any map, we say f is G-equivariant if $f(gx) = gf(x)$ for all $x \in X$ and $g \in G$.

Set 3 Problem 3:

(i) Suppose G is a Lie group acting on a C^r manifold X. Show that the maps $\mu_g : X \to X$ giveny by $\mu_g(x) = gx$ are all C^r-diffeomorphisms.

To see that μ_g is C^r for any fixed g, first note that $\iota_g : X \to G \times X$ given by $x \mapsto (g, x)$ is a C^r map. Then if $a : G \times X \to X$ is the group action, we know that $\mu_g = a \circ \iota_g$ is a composition of two C^r functions. So, μ_g is itself C^r.

Also note that μ_g is a diffeomorphism because it has the inverse function $\mu_{g^{-1}}$ which is a C^r map by identical reasoning.

(ii) As an example, show that the Lie group G acts on itself via the conjugation action $\mu_g(h) = ghg^{-1}$.

We define the $a : G \times G \to G$ by $(g, h) \mapsto \mu_g(h) = ghg^{-1}$. Then, we just need to show that the map a is smooth.

Fortunately, we know that the multiplication map $m : G \times G \to G$ given by $m(g, h) = gh$ is a smooth map. Meanwhile, if we let $\pi_1 : G \times G \to G$ be the map $\pi_1(g, h) = g$, then we know that function is smooth. Then by composing π_1 with the inverting function $i : G \to G$ which is also smooth, we get that $i \circ \pi_1 : G \times G \to G$ is a smooth map.

By the universal property of product manifolds, we now know that:

$$(g, h) \mapsto (m(g, h), i \circ \pi_1(g, h)) = (gh, g^{-1})$$

is a smooth map from $G \times G$ to $G \times G$. And now we just compose that with the multiplication map m one more time to get that $a(g, h) = ghg^{-1}$ is a smooth map.

(iii) Show that if $f : G \to H$ is a Lie group homomorphism then f has constant rank.

Note for any fixed $g \in G$ that:

$$f(h) = f(g)f(g^{-1}h) = (L' \circ f \circ L)(h).$$

where $L' : H \to H$ is the diffeomorphism given by the left multiplication of $f(g)$ and $L : G \to G$ is the diffeomorphism given by the left multiplication of g^{-1}.

But now in particular, if $h = g$ then we can see that:

$$\mathrm{d}f_g = \mathrm{d}(L' \circ f \circ L)_g = \mathrm{d}L'_e \circ \mathrm{d}f_e \circ \mathrm{d}L_g.$$

Since both $\mathrm{d}L'_e$ and $\mathrm{d}L_g$ are bijective linear maps, we can thus say that $\mathrm{d}f_g$ has the same rank as $\mathrm{d}f_e$. And since g was arbitrary, we've proven that f has constant rank equal to whatever the rank of $\mathrm{d}f_e$ is.

(iv) Suppose X, Y are two C^r manifolds both being acted on by a Lie group and $f : X \to Y$ is a G-equivariant C^1 map. If the action of G on X is transitive then f has constant rank.

Proof:
Fix any $x_0 \in X$. Then consider any other $x_1 \in X$. Because G acts on X transitively, we know there exists $g \in G$ such that $gx_0 = x_1$. And now we let $\mu_g : X \to X$ and $\widetilde{\mu}_g Y \to Y$ respectively be the maps $x \mapsto gx$ and $y \mapsto gy$.

Because μ_g is a diffeomorphism, we can say that:

$$\mathrm{d}f_{x_1} = \mathrm{d}f_{x_1} \circ \mathrm{Id}_{T_{x_1}X} = \mathrm{d}f_{x_1} \circ \mathrm{d}(\mu_g)_{x_0} \circ (\mathrm{d}(\mu_g)_{x_0})^{-1} = \mathrm{d}(f \circ \mu_g)_{x_0} \circ (\mathrm{d}(\mu_g)_{x_0})^{-1}$$

Yet then because f is G-equivariant, we know $f \circ \mu_g = \widetilde{\mu}_g \circ f$. In turn, we know that:

$$\mathrm{d}(f \circ \mu_g)_{x_0} = \mathrm{d}(\widetilde{\mu}_g \circ f)_{x_0} = \mathrm{d}(\widetilde{\mu}_g)_{f(x_0)} \circ \mathrm{d}f_{x_0}$$

So to summarize, $\mathrm{d}f_{x_1} = \mathrm{d}(\widetilde{\mu}_g)_{f(x_0)} \circ \mathrm{d}f_{x_0} \circ (\mathrm{d}(\mu_g)_{x_0})^{-1}$. Since $\mathrm{d}(\widetilde{\mu}_g)_{f(x_0)}$ and $(\mathrm{d}(\mu_g)_{x_0})^{-1}$ are both bijective linear maps, we can conclude that the rank of $\mathrm{d}f_{x_1}$ equals the rank of $\mathrm{d}f_{x_0}$. And since x_1 was arbitrary, we are done. ■

The next bit of stuff is a tangent I don't have time to do but I want to do anyways. My goal is just to illustrate that Lie groups have stronger topological properties than mere topological groups.

To start off, we have the following corollary to part (iii) of the prior problem.

Corollary: Suppose $f : G \to H$ is a Lie group homomorphism. Then f is a Lie group isomorphism if and only if f is a bijection.

Proof:
The forwards implication is trivial. As for the backwards implication, note by part (iii) of the prior problem that f has constant rank. Therefore, if f is also bijective then we know from the corollary on *page 1066* that f is a smooth diffeomorphism. And hence, f is a Lie group isomorphism. ■

Note though that we don't have an analogous result for topological groups. After all, we can give two topological groups G and H as a well as a bijective topological group homomorphism $f : G \to H$ that is not a topological group isomorphism (i.e. a group homomorphism as well as a homeomorphism).

Let $\mathbb{R}_d$ denote the real numbers with the discrete topology and let $\mathbb{R}$ denote the real numbers with the standard Euclidean topology. Both $\mathbb{R}$ and $\mathbb{R}_d$ are topological groups under addition.

Consider the map $f : \mathbb{R}_d \to \mathbb{R}$ given by $f(x) = x$. This is a bijective group homomrphism. Furthermore, f is trivially continuous. That said, f^{-1} is not continuous.

Another interesting result concerns when we know a group equipped with a topology and perhaps a smooth structure is a topological group or Lie group. But first we need two smaller results.

Lemma: Suppose G is a group equipped with a topology. Let $m : G \times G \to G$ and $i : G \to G$ denote the multiplication and inversion operations. Then G is a topological group if and only if the map $s : G \times G \to G$ given by $s(g, h) = gh^{-1}$ is continuous.

Proof:
For $i = 1, 2$ we let $\pi_i : G \times G \to G$ be the projection $(g_1, g_2) \mapsto g_i$. Both of these are continuous maps by the definition of the product topology.

$(\Longrightarrow)$
If G is a topological group then $s = m \circ (\pi_1, i \circ \pi_2)$ is a composition of continuous maps and is thus continuous.

$(\Longleftarrow)$
To start off, note that the map $\iota_e : G \to G \times G$ given by $\iota_e(g) = (e, g)$ is continuous. Also, $i = s \circ \iota_e$. So, i is a continuous map. Next, note that $m = s \circ (\pi_1, i \circ \pi_2)$ is a composition of two continuous functions and is thus continuous.

Furthermore, if G is a smoth manifold then G is a Lie group if and only if the map $s : G \times G \to G$ given by $s(g, h) = gh^{-1}$ is smooth.

The proof is literally identical to before. Just swap out the word "continuous" with the word "smooth". ■

The other result we need is an exercise from John Lee.

Problem 7-1: Suppose G is a group that is also a smooth manifold. Also suppose the group map $m : G \times G \to G$ given by $m(g, h) = gh$ is smooth. Then m is a submersion.

Proof:
To start off, we only need the smoothness of m to prove that the map $R_g(h) = gh$ is a C^∞ diffeo-automorphisms for all $g \in G$. And thus by identical reasoning as on *page 1233*, we can conclude that G will have no boundary. This also in turn means that $G \times G$ has no boundary. And now in order to prove that m is a submersion it is equivalent due the local section theorem on *pages 1106-1107* to show that for any fixed $(g_1, g_2) \in G \times G$ there exists an open set $U \subseteq G$ containing $g_1 g_2$ and a smooth map $\sigma : U \to G \times G$ such that $\sigma(g_1 g_2) = (g_1, g_2)$.

This is easy though and the σ we will construct is even a *global* section. Just define $\sigma : G \to G \times G$ by $\sigma(g) = (gg_2^{-1}, g_2)$ Then σ is smooth, $\sigma(g_1 g_2) = (g_1, g_2)$, and $m \circ \sigma(g) = g$ for all $g \in G$. ■

Now we're ready for the actual interesting result (which is also an exercise from John Lee).

Problem 7-3: Suppose that G is a group that is also a smooth manifold. Also, let $m : G \times G \to G$ and $s : G \times G \to G$ be the maps $m(g, h) = gh$ and $s(g, h) = gh^{-1}$. If m is a smooth map then so is s. Then consequently by our prior lemma, we know G is a Lie group.

Proof:
Define $F : G \times G \to G \times G$ by $F(g, h) = (g, hg)$. Also for $i = 1, 2$ let $\pi_i : G \times G \to G$ be the map $(g_1, g_2) \mapsto g_i$.

My first obervation is that F is a smooth map.

After all, F can be written as $F = (\pi_1, \ \ m \circ (\pi_2, \pi_1))$.

Secondly, I claim that F is a local diffeomorphism.

Since G and thus $G \times G$ has no boundary (see my proof of the last problem), it suffices by the inverse function theorem to prove that $\mathrm{d}F_(g, h)$ has full rank at all $(g, h) \in G \times G$. The easiest way to go about this is to prove that $\mathrm{d}F_{(g,h)}$ is injective for all $(g, h) \in G \times G$.

We start by making the standard identification $T_{(g,h)}(G \times G) \cong T_g G \oplus T_h G$. Then note for any $(v, w) \in T_g G \oplus T_h G$ that:

$$\begin{aligned} \mathrm{d}F_{(g,h)}(v, w) &= (\mathrm{d}(\pi_1)_{(g,h)}(v, w), \ \ \mathrm{d}(m \circ (\pi_2, \pi_1))_{(g,h)}(v, w)) \\ &= (v, \ \ \mathrm{d}m_{(h,g)}\,(\mathrm{d}(\pi_2)_{(g,h)}(v, w), \ \mathrm{d}(\pi_1)_{(g,h)}(v, w))) \\ &= (v, \ \ \mathrm{d}m_{(h,g)}\,(w, v)). \end{aligned}$$

It easily follows that $(v, w) \in \ker(\mathrm{d}m_{(g,h)})$ if and only if $v = 0$ and $(w, 0) \in \ker(\mathrm{d}m_{(h,g)})$

Next let $R_g : G \to G$ be the map $R_g(h) = hg$. We know that R_g is a diffeomorphism (see the prior problem). Also note that $R_g = m \circ (\mathrm{Id}_G, C_g)$ where $C_g : G \to G$ is the constant map sending everything to g. As a result, for any $w \in T_h G$ we have that:

$$\mathrm{d}(R_g)_h w = \mathrm{d}m_{(h,g)}(\mathrm{d}(\mathrm{Id}_G)_h w, \ \mathrm{d}(C_g)_g w) = \mathrm{d}m_{(h,g)}(w, 0).$$

But because R_g is a diffeomorphism, we know that $\mathrm{d}(R_g)_h w = 0$ if and only if $w = 0$. So, $(w, 0) \in \ker(\mathrm{d}m_{(h,g)})$ implies that $w = 0$. And with that, we've finished proving that the kernel of $\mathrm{d}F_{(g,h)}$ is trivial.

Thirdly I observe that F is a bijection whose inverse function is $F^{-1}(g, h) = (g, hg^{-1})$.

The three prior facts combined let us conclude that F is a smooth diffeomorphism from $G \times G$ to $G \times G$. In other words, we now know that F^{-1} is a smooth function.

Finally, we can now conclude that s is smooth since:

$$s = \pi_2 \circ F^{-1} \circ (\pi_2, \pi_1). \ ■$$

There is not an analogous result when we are only working with topological groups.

For example, let $\mathbb{R}_l$ be the set of real numbers equipped with the lower limit topology

(i.e. the topology generated by the basis of all half intervals $[a,b)$). We can consider $\mathbb{R}_l$ a group under addition.

I shall first show that the map $a:\mathbb{R}_l\times\mathbb{R}_l\to\mathbb{R}_l$ given by $a(x,y):=x+y$ is continuous.
It suffices to show for any $(x,y)\in\mathbb{R}_l\to\mathbb{R}_l$ and open set $U\subseteq\mathbb{R}_l$ cotaining $x+y$ we have that there is an open set $V\subseteq\mathbb{R}_l\times\mathbb{R}_l$ containing (x,y) with $a(V)\subseteq U$. Fortunately, by shrinking U we can always without loss of generality assume $U=[x+y,x+y+2\varepsilon)$ for some $\varepsilon>0$. And now we just set:
$$V=[x,x+\varepsilon)\times[y,y+\varepsilon).$$

That said, also observe for any sequence $\{x_k\}_{k\in\mathbb{N}}\subseteq\mathbb{R}_l$ that $x_k\to x$ in $\mathbb{R}_l$ iff $x_k\to x$ in the standard Euclidean topology and there exists $K\in\mathbb{N}$ such that for all $k\geq K$ we have that $x_k\geq x$. Using this fact, we can easily show that the negation map $n:\mathbb{R}_l\to\mathbb{R}_l$ given by $n(x)=-x$ is *not* continuous.
To see why, note that $\frac{1}{k}\to 0$ in $\mathbb{R}_\ell$ by $n(\frac{1}{k})=\frac{-1}{k}\not\to 0$ in $\mathbb{R}_\ell$.

Therefore, $\mathbb{R}_l$ is *not* a topological group even though its addition operation is continuous.

The next notes finally return to what I was doing on *page 1157*.

My end goal before getting distracted by singular homology was to show that most C^r manifolds we would care about can be embedded into Euclidean space if they weren't at first, and in fact if M is n-dimensional then it can be embedded into $\mathbb{R}^{2n+1}$. To accomplish this goal, it suffices to show that:

1. We can embed M into $\mathbb{R}^N$ for just some N.
2. Given an n dimensional embedded submanifold $M\subseteq\mathbb{R}^N$ where $N>2n+1$, there is an embedding of M into $\mathbb{R}^{N-1}$.

Lemma 6.13 on *pages 1156-1157* takes a step towards addressing the second task, albeit the lemma only gives us an immersion and not an embedding of M. At least in the case though that M is compact, then proposition 4.22 on *page 1104* tells us that our injective immersion is automatically an embedding. So for compact manifolds we are half done.

Lemma: Suppose M is a compact n-dimensional C^r manifold (where $r\geq 1$). Then there exists $N\in\mathbb{N}$ and a C^r embedding $f:M\hookrightarrow\mathbb{R}^N$.
Proof:
For every $p\in M$ we can pick a smooth chart (U_p,φ_p) such that $p\in U_p$. And then we can pick two more open sets W_p and V_p such that $p\in W_p\subseteq\overline{W_p}\subseteq V_p\subseteq\overline{V_p}\subseteq U_p$. But now as M is compact, we know there are finitely many $p_1,\ldots,p_\ell$ such that $M\subseteq\bigcup_{i=1}^{\ell}W_{p_i}$.

For each i let $b_i:M\to[0,1]$ be a C^r bump function with $b_i\equiv 1$ on $\overline{W_{p_i}}$ and $b_i\equiv 0$ on $M-V_{p_i}$. Then by pasting lemma, we can guarentee that the product $b_i\varphi_{p_i}$ defines a C^r map from M to $\mathbb{R}^n$ for each i (where the product equals 0 at all points outside the

domain of the chart (U_{p_i}, φ_{p_i})). So, we now define the map $f : M \to \mathbb{R}^{(n+1)\ell}$ by:

$$f(p) \coloneqq (b_1(p), \ldots, b_\ell(p), (b_1\varphi_1)(p), \ldots, (b_\ell\varphi_\ell)(p))$$

Because M is compact, in order to prove that f is a C^r embedding we just need to explain why f is a C^r injective immersion.

Fortunately, because every component function of f is a C^r map, we automatically have that f is a C^r map as well.

Also suppose $f(p) = f(q)$. Since $M \subseteq \bigcup_{i=1}^{\ell} W_{p_i}$ we know that $p \in W_{p_i}$ for some i. In turn $b_i(p) = 1$, and since $f(p) = f(q)$ this also guarentees that $b_i(q) = 1$. So, both p, q are covered by the chart (U_{p_i}, φ_{p_i}) and $\varphi_i(p) = (b_i\varphi_i)(p) = (b_i\varphi_i)(q) = \varphi_i(q)$. Finally, since φ_{p_i} is injective on U_{p_i} and $p, q \in U_{p_i}$, we can conclude that $p = q$. So, we have shown that f is injective.

Finally, to prove that f is an immersion, consider any $p \in M$ and then pick i with $p \in W_{p_i}$. Importantly, $b_i \equiv 1$ on a neighborhood of p. And now upon identifying:

$$T_{f(p)}\mathbb{R}^{(n+1)\ell} \cong \left(\bigoplus_{i=1}^{\ell} T_{b_i(p)}\mathbb{R}\right) \oplus \left(\bigoplus_{i=1}^{\ell} T_{(b_i\varphi_i)(p)}\mathbb{R}^n\right)$$

we have that:

$$\mathrm{d}f_p = (\mathrm{d}b_1, \ldots, \mathrm{d}b_\ell, \mathrm{d}(b_1\varphi_{p_1})_p, \ldots, \mathrm{d}(b_{i-1}\varphi_{p_{i-1}})_p, \mathrm{d}(\varphi_i)_p, \mathrm{d}(b_{i+1}\varphi_{p_{i+1}})_p, \ldots, (b_\ell\varphi_{p_\ell})_p).$$

Yet $\mathrm{d}(\varphi_i)_p : T_pM \to T_{\varphi_i(p)}\mathbb{R}^n$ is a linear bijection (and thus has rank $n = \dim(M)$). It then follows that $\mathrm{d}(f)_p$ must also have rank at least $n = \dim(M)$. So, $\mathrm{d}f_p$ is full rank and thus injective. And since p was arbitrary, we've finished proving that f is an immersion. ■

Corollary (Whitney embedding theorem for compact manifolds): Suppose M is a compact n-dimensional C^r manifold (where $r \geq 1$). Then there exists $N \in \mathbb{N}$ and a C^r embedding $f : M \hookrightarrow \mathbb{R}^N$. If in addition we assume $r \geq 2$, then we can specifically choose N to be equal to $2n + 1$.

The case of noncompact C^r manifolds is more delicate and in fact Oprea skips going over this case in lecture. I don't want to skip this though.

To start off, suppose X is a topological space. We say a function $f : X \to \mathbb{R}$ is an exhaustion function if for every $c \in \mathbb{R}$ the set $f^{-1}((-\infty, c])$ is compact.

Theorem 2.28: Suppose M is a C^r manifold (where $r \geq 1$). Then M admits a postive C^r exhaustion function $f : M \to (0, \infty)$.

Proof:

If M is already compact, then we can just pick f to be a constant function. So without loss of generality we will now assume M is not compact.

Using the fact that M is second countable as well as an LCH space, we can find a sequence $\{U_k\}_{k\in\mathbb{N}}$ of precompact open sets on M such that $M \subseteq \bigcup_{k=1}^{\infty} U_k$. Next, we can pick C^r partition of unity $\{\phi_k\}_{k\in\mathbb{N}}$ subordinate to the sets U_k. In other words, ϕ_k is positive and C^r for all k, $\text{supp}(\phi_k) \subseteq U_k$ for all k, $\sum_{k=1}^{\infty} \phi_k = 1$, and every $p \in M$ has a neighborhood on which only finitely many ϕ_k are ever nonzero.

Finally, we define $f \coloneqq \sum_{k=1}^{\infty} k\phi_k$. Note that f is a C^r map because locally around any $p \in M$ we have that f is just a linear combination of finitely many of the ϕ_K. Also note that $f(p) > 0$ for all $p \in M$. After all, $\phi_{k'}(p) > 0$ for some k'. And then since all the summands are nonnegative, we can conclude that $f(p) > k'\phi_{k'}(p) > 0$.

As for showing that f is an exhaustion function, consider any $c \in \mathbb{R}$. We can pick a positive integer $N > c$. And now if $p \notin \bigcup_{i=1}^{N} U_{p_i}$ we know that:

$$\phi_1(p) = \phi_2(p) = \cdots = \phi_N(p) = 0.$$

As a result: $f(p) = \sum\limits_{k=N+1}^{\infty} k\phi_k(p) > N \sum\limits_{k=N+1}^{\infty} \phi_k(p) = N \cdot 1.$

This proves that $f^{-1}((-\infty, c]) \subseteq f^{-1}((-\infty, N]) \subseteq \bigcup_{i=1}^{N} U_{p_i} \subseteq \bigcup_{i=1}^{N} \overline{U_{p_i}}$. Since the right most set is a union of compact sets and thus compact itself, and also because $f^{-1}((-\infty, c])$ is a closed subset of M due to f being continuous, we can thus conclude that $f^{-1}((-\infty, c])$ is compact. ■

Let $\lambda(k)$ denote the number of time continuously differentiable we need a function $f : \mathbb{R}^{n+k} \to \mathbb{R}^n$ to be to prove that Sard's theorem holds for f.

See *page 1154* to recall what's up with $\lambda(k)$. (Namely, $\lambda(k) = k + 1$ for all $k \geq 0$. That said, the lower bound which *I* managed to prove on $\lambda(k)$ is weaker.)

Recall from two pages ago the two tasks we need to accomplish. We'll accomplish the first task for a noncompact manifold by using a C^r exhaustion function, Sard's theorem, and the Whitney embedding theorem for compact manifolds that we already proved.

Theorem 6.15: Suppose M is a noncompact n-dimensional C^r manifold and that $r \geq \max(2, \lambda(n-1))$ (which equals $\max(2, n)$). Then there is a C^r embedding $F : M \hookrightarrow \mathbb{R}^{4n+3}$.

Proof:
Let $f : M \to (0, \infty)$ be a C^r exhaustion function. Because $r \geq \lambda(n-1)$, we know that the set of critical values of f in $(0, \infty)$ has measure zero. Hence, for every nonnegative integer i we can pick real numbers $i < a_i < b_i < i + 1$ such that a_i and b_i are regular values of f.

Now define:

- $D_0 \coloneqq f^{-1}((-\infty, 1])$ and $E_0 \coloneqq f^{-1}((-\infty, a_1])$,
- $D_i \coloneqq f^{-1}([i, i+1])$, and $E_i \coloneqq f^{-1}([b_{i-1}, a_{i+1}])$ for all $i \geq 1$.

By proposition 5.47 on *pages 1127-1128*, we can conclude that each E_i is a compact regular domain of M for all $i \geq 0$. Furthermore, each D_i will be a compact subset of M contained within the interior of each E_i.

For all $i \geq 0$ we let $\phi_i : M \to [0,1]$ be a C^r bump function such that $\phi_i \equiv 1$ on a neighborhood of D_i and $\operatorname{supp}(\phi_i) \subseteq E_i^\circ$. Also, for all $i \geq 0$ we let $g_i : E_i \hookrightarrow \mathbb{R}^{2n+1}$ be a C^r embedding.

We know such an embedding exists due to Whitney's embedding theorem for compact manifolds as well as the fact that $r \geq 2$.

Importantly, the product $\phi_i g_i$ then defines a C^r map on all of M (where $(\phi_i g_i)(p) = 0$ if $p \notin E_i$.) Also observe that every point $p \in M$ has a neighborhood U intersecting at most two of the E_i. In fact, we can even even say that U intersects E_i and E_{i+1} for some i.

To see, why note that $f^{-1}((-\infty, a_1))$ intersects only E_0 and E_1. Also, for any $i \geq 0$ we have that $f^{-1}((a_i, a_{i+1}))$ and $f^{-1}((b_i, b_{i+1}))$ intersect only E_i and E_{i+1} on a neighborhood of p.

Now we define $F : M \to \mathbb{R}^{4n+3} \cong \mathbb{R}^{2n+1} \times \mathbb{R}^{2n+1} \times \mathbb{R}$ by:

$$F(p) \coloneqq \left(\sum_{i \text{ even}} (\phi_i g_i)(p), \quad \sum_{i \text{ odd}} (\phi_i g_i)(p), \quad f(p) \right)$$

Note that F is a C^r map because every $p \in M$ has a neighborhood on which only one summand from each of the sums in the definition of F is nonzero. In a similar vein, to see that F is an immersion, note for any $p \in M$ that there exists i with $p \in D_i$. In turn, $(\phi_i g_i) = g_i$ on a neighborhood of p. Then combining the fact that $\mathrm{d}(g_i)_p$ is injective with the observation of what the formula for F locally looks like will prove $\mathrm{d}F_p$ is injective.

As for showing that F is injective, note that if $F(p) = F(q)$ then we know that $f(p) = f(q)$. In particular, this means that if $p \in D_i$ then $q \in D_i$ also. So, $\phi_i(p) = 1 = \phi_i(q)$. Next note that $F(p) = F(q)$ implies that:

$$g_i(p) = (\phi_i g_i)(p) = (\phi_i g_i)(q) = g_i(q).$$

Yet $p, q \in D_i$ implies that $p, q \in E_i$ as well. And since g_i is injective on E_i, we can conclude that $p = q$.

Finally, I claim that F is a proper map. To see why, suppose $K \subseteq \mathbb{R}^{4n+3}$ is any compact set. Then in particular, there must exist $A, B \in \mathbb{R}$ such that $K \subseteq \mathbb{R}^{2n+2} \times [A, B]$. Hence, $F^{-1}(K) \subseteq f^{-1}([A, B])$. Yet now note that $F^{-1}(K)$ is a closed set. Meanwhile, f being an exhaustion function means that $f^{-1}([A, B])$ is a compact set. It follows that $F^{-1}(K)$ is compact.

By proposition 4.22 on *page 1104*, we can conclude that F is a C^r embedding. ■

As for accomplishing the second task we first define for any 1-dimensional linear subspace $S \subseteq \mathbb{R}^N$ and any $R > 0$ the tube with axis S and radius R given by:

$$T_R(S) \coloneqq \{x \in \mathbb{R}^N : |x - y| < R \text{ for some } y \in S\}.$$

Note that if $S = \operatorname{span}(v)$ then $\min_{y \in S} |x - y| = \left| x - \frac{x \cdot v}{|v|^2} v \right|$.

Importantly, the map $x \mapsto \left| x - \frac{x \cdot v}{|v|^2} v \right|$ is a continuous function and so we can conclude that $T_R(S)$ is an open set.